Emilio Palencia

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Education

Toronto Metropolitan University — Toronto, ON
Bachelor of Engineering, Mechanical Engineering (Co-op) — *Sep. 2025 – Apr. 2030 (Expected)*

Experience

Mechanical Design Team Member — July 2026 – Present
Toronto MetRobotics — *Toronto, ON*

- Designed a SolidWorks wrist-mounted end effector for a rover manipulator to fit a specified mounting interface, motor package, and 0.10–1.00 kg payload range; produced GD&T drawings and iterated through three PLA prototypes, increasing wall thickness to 4 mm after FEA evaluation of deformation under a 1.5 kg test load.
- Sized a NEMA 17 stepper (0.45 N·m holding torque) with a 24:1 worm reduction for a 4-claw gripper; required 0.74 N·m for a 1.0 kg payload; calculated 10.8 N·m ideal and 4.32 N·m output at an assumed 40% worm efficiency (safety factor 5.8)

Structural Design Lead — Sep. 2025 – Nov. 2025
Toronto Metropolitan University Project Design — *Toronto, ON*

- Led the structural workstream for a 30 m enclosed pedestrian skybridge feasibility study (team of 6); compared Warren and Pratt truss layouts and selected Warren; documented hazards, materials, and accessibility and sustainability requirements.
- Built a cost and economic model (cost benchmarking, break-even sensitivity, 30-year NPV/IRR) on an assumed $5.5M capital cost and estimated pedestrian time savings, yielding an 8.1% simple annual return and a positive 30-year NPV.

Payment Processing Specialist — Jan. 2025 – Mar. 2026
SinglePoint GI Inc. — *Toronto, ON*

- Resolved billing discrepancies across 60+ daily client interactions while maintaining 95%+ customer-satisfaction performance.

Projects

Drill System Comparison Study | *Ansys, Python* — Aug. 2026 – Present

- Comparing tricone and PDC bit models in Ansys Workbench under 50–100 kN weight-on-bit and assumed torque coefficients (0.25 tricone / 0.50 PDC) across four rock cases (granite, sandstone, permafrost, saprolite; UCS 10–300 MPa); analysis in progress.
- Built a PyQt application (NumPy/SciPy) that solves Teale's specific-energy equation for penetration rate, with torque estimated from weight-on-bit and a UCS-based drilling-efficiency target, to compare simulated penetration rate, drilling time, rotary power, and energy consumption across operating conditions, rock strengths, and bit torque coefficients.

5-DOF Robotic Arm | *SolidWorks, Python, Arduino* — May 2026 – July 2026

- Designed and 3D-printed a 5-DOF robotic arm driven by 4 MG996R servos at the joints and a mirrored MG996R pair at the base for more torque and balanced load, with a hand-held potentiometer master controller (ESP32) teleoperating the Arduino-controlled slave arm via Bluetooth; manually calibrated position inputs.
- Developed a desktop control application (tkinter) for manual operation, calibration, and motion recording/playback, allowing recorded operator movements to be replayed by the arm.

Technical Skills

CAD & Design: SolidWorks (Certified SolidWorks Associate, CSWA), Fusion 360, AutoCAD, GD&T
Analysis: Ansys Workbench, Finite Element Analysis
Programming & Embedded: Python, C, Arduino, ESP32, Git, MATLAB/Simulink
Fabrication & Safety: 3D printing (PLA, PETG), WHMIS, Machining Fundamentals & Safety Awareness
Languages: English (Native), Spanish (Native), French (Intermediate)
Other Skills: Microsoft Office 365, Google Workspace